

Semantic Ifs over Images: Logit-Readout Decisions with a Vision–Language Model

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<https://github.com/XxSamaxX/discern>

Abstract

Recent open implementations read a decision directly from a language model’s answer-token logits instead of decoding text, turning a model into a runtime-defined conditional—a *semantic if*. We port this readout unchanged to a vision–language model and evaluate it on MMBench (en/dev, 4329 items) plus a small hand-built probe. Qwen3-VL-4B reaches 87.4% with one forward pass per item at 91 ms, and 82.2% under MMBench’s CircularEval protocol, which rotates the answer options.

Our main results concern the confidence signal rather than the score. The probability these systems report and the logit gap behind it are not competing signals: the probability is a monotone transform of the gap, exactly so on binary questions, and the two predict correctness equally well (AUC 0.895 against 0.894). The difference is representational and decisive in practice. Past 17.33 nats the probability is no longer representable in float32, the dtype most inference pipelines carry, and on MMBench that covers 62% of all items, where every probability is exactly 1.0; within that collapsed set the gap still separates correct from incorrect at AUC 0.736. Thresholding the gap gives a usable operating point—16 nats answers 66% of items at 98.4% accuracy against an 87.4% base rate—and the same threshold expressed as a probability does not survive the dtype. The gap also predicts order sensitivity almost perfectly (AUC 0.980): rotating options changes the winner on 11.7% of items, and those items have a median gap of 2.7 nats against 22.8 for the stable ones, which bounds when order calibration is worth its n forward passes.

On POPE (9000 binary presence questions) the same readout scores 89.6%, but the AUC of both forms of the signal falls to 0.777 and the median gap when the model is wrong rises from 5.2 to 16.4 nats: with two slots the model is not merely wrong but confidently wrong, and a threshold tuned on four-way questions does not transfer to two-way ones.

Two negative results survive the scale-up. First, difficulty is not epistemics: on 12 normative questions no model can ground, the gap distribution shifts (median 8.8 vs 20.9 nats, AUC 0.788, $p = 0.006$) but overlaps so heavily that the best threshold still admits 7 of 12—the model asserts that a photograph requires legal consent at 15.6 nats, as confidently as it reports that no dog is present at 26.4. Second, our own 22-item probe scored 22/22 with every softmax probability above 0.9993, which we initially read as universal saturation; the 4329-item run shows that this was a property of an over-easy benchmark, and we report it as a caution about small author-written test sets.

1 Introduction

A recurring pattern in recent agent tooling is to stop asking a language model for text and ask it for a decision. The construction is simple: phrase the decision as a lettered multiple choice, run a single forward pass, and read the logits at the final position restricted to the tokens A, B, C. There is no decoding loop, so the cost is one prefill. The open SemIf project [3] publishes this readout as a baseline, and the JEV-CPU port [6] demonstrates it on a laptop CPU with a 0.6B model. Practitioners describe the result as a *semantic if*: a conditional whose predicate is written in natural language at runtime rather than compiled.

The attraction for computer vision is immediate. A detector trained on a fixed label set answers “is there a person” but cannot answer “is this photograph posed”, “is the white balance wrong”, or “is the meal already in progress”. These are not classes; they are predicates. If the readout transfers to a vision–language model, the predicate becomes free text and no training set is required.

This paper reports what happens when it does transfer. Our contributions are:

1. A port of the readout to a vision–language model in which the scoring path is byte-identical to the text implementation; only prompt construction and model loading differ (Section 3).
2. An evaluation on MMBench en/dev: 87.4% at 91 ms per item, 82.2% under CircularEval (Section 4.5), plus a 22-item adversarial probe the model answers in full (Section 4.1).
3. A correction to a claim that is easy to make and that we made ourselves: the reported probability and the logit gap are one signal, not two, and discriminate identically. What separates them is float32, in which the probability collapses to exactly 1.0 on 62% of items while the gap still ranks them at AUC 0.736. Thresholding the gap supports abstention at 98.4% accuracy over 66% of items (Sections 4.2 and 4.5).
4. A characterisation of position bias: 1–2 nats in magnitude, flipping the winner on 11.7% of items, with the gap predicting order-stability at AUC 0.980. This bounds when order calibration earns its n forward passes (Section 4.3).
5. A negative result we consider the most consequential for deployment: the gap does not separate answerable perceptual questions from ungroundable normative ones well enough to gate them (Section 4.4).
6. A methodological caution, learned the hard way: our own 22-item benchmark reported universal softmax

saturation and zero order sensitivity, and the 4329-item run refuted both.

2 Related work

Open-vocabulary detection attacks the same limitation from the other side. CLIP-style image-text alignment [8] underpins detectors such as OWL-ViT [7], Grounding DINO [5] and YOLO-World [1], which accept a free-text vocabulary and localise its members. These systems remain far faster than a vision-language forward pass and they return boxes, which the readout studied here does not. They are, however, restricted to naming things: they cannot evaluate a predicate over a scene. The two approaches are complementary rather than competing, and a natural composition is to let a detector propose regions and a semantic if evaluate predicates on the crops.

Our negative results connect to two literatures. Position bias in multiple choice prompting is well documented [11], and contextual calibration [10] is the standard remedy; we measure the bias in nats and find it an order of magnitude smaller than the decision margins on perceptual items, which bounds when that remedy is worth its cost. Separately, modern networks are known to be poorly calibrated [2]; the failure we document is not one of calibration at all, but of numeric range—the reported probability orders items exactly as well as the logits beneath it, and loses that order only when stored at the precision these pipelines use. Finally, object hallucination benchmarks such as POPE [4] establish that vision-language models assert the presence of absent objects; our false-premise and absence items probe the same behaviour and, notably, do not reproduce it at this model scale.

3 Materials and methods

3.1 The decision readout

A decision is a tuple of evidence, a criterion, and 2–16 options. The prompt is a JSON payload naming each option with a letter, under a system instruction to answer with that letter alone. The implementation verifies that each letter encodes to exactly one token with an exact decode round trip, and that appending the letter does not alter the tokenisation of the prompt that precedes it; both checks raise rather than silently degrade. A single forward pass with `use_cache=False` yields the final-position logit vector, which is indexed at the answer-slot tokens. A softmax over those few entries produces the reported probabilities.

3.2 Extension to images

We changed exactly two things. The user turn carries an image part before the JSON payload, and the model is loaded as an image-text-to-text model in bfloat16 on the GPU. The slot verification, the forward call and the logit indexing are unchanged from the text implementation, so results are directly comparable. At the processor’s default resolution a 640×480 photograph expands to roughly 300

visual tokens, giving prompts of 351–413 tokens in our benchmark.

3.3 Benchmark

We wrote 22 decisions over four photographs from COCO *val2017*. Twelve are ordinary perceptual judgements, three of which a fixed-vocabulary detector could also answer and act as controls. Ten target families known to be difficult for vision-language models: exact counting, absence, a false-premise question (the colour of a car in an image containing no car), an unanswerable question (the time of day in a windowless interior), fine-grained comparison, and spatial relation. Ground truth was written by the author from the images. One label was initially wrong—we judged a brand mark illegible, and the model read it correctly—and was corrected after visual inspection of an enlarged crop; we report this because the corrected item is the hardest in the set.

A second set of 12 *normative* questions was written to be structurally identical but institutionally ungrounded: whether consent must be obtained, whether a retention policy requires deletion, whether a model release has expired. Each carries one sentence of evidence and two options. No model can answer these from the evidence given, because the answer depends on a policy it has never seen.

3.4 Order calibration and the logit gap

For each decision we optionally re-score under all cyclic rotations of the option list and average in logit space. We record, per decision, the gap in nats between the two highest mean logits, whether the winner changes across rotations, and the mean logit received by each letter slot, which with rotation is a clean estimate of position bias.

3.5 Hardware and models

All experiments run on one machine: an Intel i9-12900F (8 performance and 8 efficiency cores) with 31 GiB of RAM and an NVIDIA RTX 3080 with 12 GiB. The vision model is Qwen3-VL-4B-Instruct in bfloat16 on the GPU. The text models are Qwen3-0.6B [9] and Qwen3.5-4B in float32 on the CPU; the latter is the model the upstream manifest [3] designates for this readout, and the CPU port substitutes the former.

4 Results

4.1 Perceptual accuracy

Qwen3-VL-4B answers 22 of 22 correctly. All four adversarial families are handled: it counts two cats and two remote controls, denies the presence of a dog and of a cat in the rooms that contain neither, answers “there is no car” to the false-premise question at a gap of 22.1 nats, and answers “cannot be determined” to the unanswerable one at 15.5 nats. It reads an Adidas mark on a cap occupying roughly 20×15 pixels of a 640×480 image. The

two lowest gaps in the set are the fine-grained comparison (7.3 nats) and the spatial relation (8.1 nats), which are the two items we judged hardest a priori. Throughput is 110–140 ms per decision at 8.4 GiB of VRAM, after a first forward pass of 11.2 s attributable to CUDA warm-up.

4.2 The probability and the gap are one signal in two scales

It is tempting to say that these systems report a useless probability and that the logit gap is the real signal. We tested that and it is false. The softmax over the answer slots is a monotone transform of the gap: on POPE’s binary questions it is exactly $p = \sigma(\text{gap})$, with zero monotonicity inversions across 8999 items, and on MMBench’s three- and four-way questions only 4.5% of adjacent pairs invert. As ranking signals they are the same quantity. Their AUCs for predicting correctness agree to the third decimal: 0.895 for the probability against 0.894 for the gap on MMBench, and 0.777 for both on POPE. A threshold on one is a threshold on the other—requiring $p \geq 0.99$ retains 286 of 547 MMBench errors, and requiring a gap of 4.60 nats, its exact equivalent, retains 288.

The difference is representational, and in practice it is decisive (Figure 1). The probability compresses the entire useful range into the last few digits before one: a gap of 12 nats is $p = 0.99999386$, 20 nats is $p = 0.99999998$, and past $-\ln 2^{-25} = 17.33$ nats the value is no longer representable in float32, the dtype most inference pipelines carry. On MMBench that is **62% of all items**, where every probability is exactly 1.0 and the ranking is gone; in float16, whose collapse threshold is only 8.32 nats, it is 81%. Within that collapsed set the gap still separates correct from incorrect answers at AUC 0.736, and in the float16 set at 0.867. That is information the readout computed and the representation threw away.

The practical recommendation therefore survives, for a narrower reason than we first believed: compute, log and threshold the gap in nats. Not because it discriminates better—it does not—but because it is the only one of the two that survives contact with a float32 pipeline. The upstream implementation labels its output “*conditional option score; uncalibrated as decision confidence*”, which is correct; we would add that the number is also the wrong scale on which to store it.

4.3 Order sensitivity is small

Averaged over rotations, the mean logit assigned to each letter slot differs by 0.2–2.4 nats, with no consistent preference for the first slot. Against perceptual gaps of 7.3 nats and above, this is an order of magnitude too small to matter, and indeed the winner changes under rotation in 0 of 22 perceptual decisions. Order calibration on this workload costs n forward passes and buys nothing. It is not free elsewhere: in the text-model experiments of Section 4.7, where gaps fall to 0.8–2.3 nats, the same 1–2 nats of position bias decides the answer outright and the winner flips. A practical rule follows: rotate and average only when the single-pass gap is below roughly 3 nats.

Table 1: Qwen3-VL-4B on MMBench en/dev (4329 items). CircularEval requires the correct answer under every rotation of the options.

Protocol	Accuracy	ms/item	AUC gap
VanillaEval	87.4%	91	0.894
CircularEval	82.2%	347	0.956

4.4 The gap does not gate normative questions

If the gap tracks difficulty within a family, the natural hope is that it also flags questions the model has no business answering. We tested this on the 12 normative items, which are unanswerable in principle rather than merely hard.

The hope fails (Figure 2). The distributions do differ: median 8.8 nats for normative against 20.9 for perceptual, giving an AUC of 0.788 (Mann–Whitney $U = 208$, $z = 2.74$, $p = 0.006$). But 7 of 12 normative items fall inside the perceptual range, and the best achievable threshold, 7.3 nats, retains all 22 perceptual decisions only by admitting 7 of the 12 normative ones—79% accuracy for a filter whose entire purpose is to catch the remainder. Individual cases are worse than the aggregate suggests. The model asserts that a photograph with a colour cast should enter the correction queue at 23.1 nats and that publication must be held for consent at 15.6, both policy questions, at gaps indistinguishable from its report that there is no dog in the picture (26.4). Only 1 of 12 normative items flips under rotation.

Two of our normative items are arguably entailed by their evidence, which inflates their gaps and weakens the contrast; we note this rather than remove them post hoc.

4.5 Validation at scale on MMBench

Our own benchmark returned 22/22, so it could not measure whether the gap *predicts* anything—there were no errors to predict. MMBench supplies 547 of them under VanillaEval.

The gap separates correct from incorrect answers with AUC 0.894 ($z = 29.9$), with a median of 22.4 nats when the model is right against 5.2 when it is wrong. Thresholding on it yields a genuine abstention policy: at 12 nats the system answers 75% of items at 96.8% accuracy, and at 16 nats 66% at 98.4%, against an 87.4% base rate (Figure 3). The same threshold expressed as a probability selects the same items (Section 4.2); what it does not do is survive being stored, since 62% of these items share the single float32 value 1.0.

CircularEval costs 5.2 points (87.4% to 82.2%) because the winner changes under rotation on 507 of 4329 items, 11.7%. Those items are not randomly distributed: their median gap is 2.7 nats against 22.8 for the order-stable ones, and the gap predicts stability with AUC 0.980 (Figure 4). The rule we proposed in Section 4.3 from a 22-item probe—rotate only below roughly 3 nats—was written before this dataset was downloaded and lands on the empirical median of the items that actually flip.

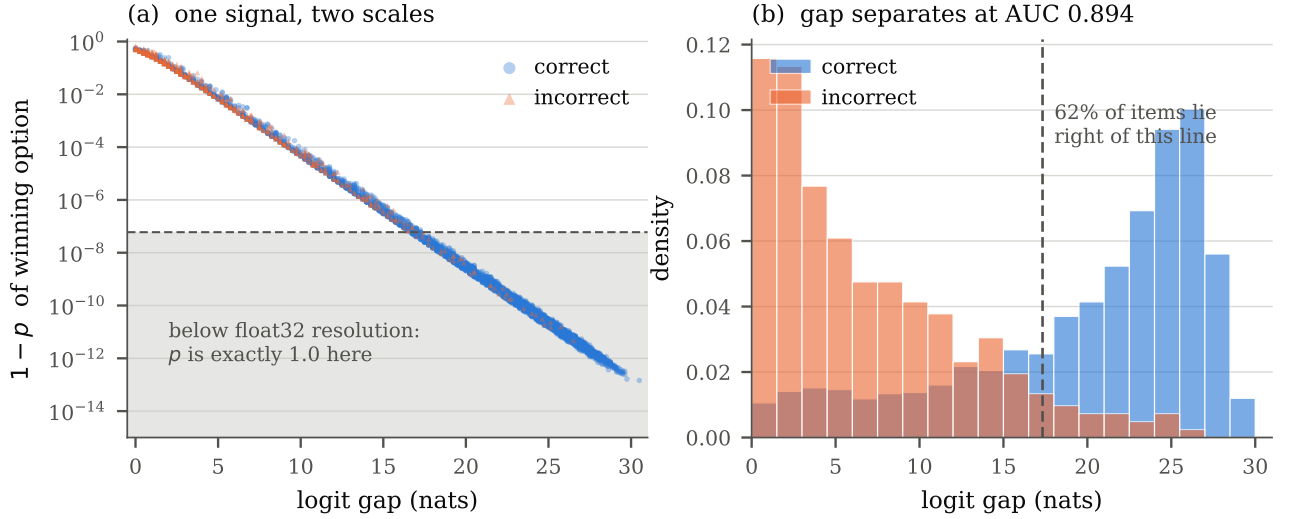


Figure 1: 4329 MMBench items. **(a)** The reported probability is a monotone transform of the logit gap—the relation is a line, and on binary questions it is exactly $p = \sigma(\text{gap})$ with zero inversions in 8999 POPE items. The shaded band is where $1 - p$ falls below float32 resolution and the probability becomes exactly 1.0. **(b)** Gap distributions for correct and incorrect answers. 62% of items sit right of the dashed line, where a float32 probability can no longer tell them apart and the gap still can.

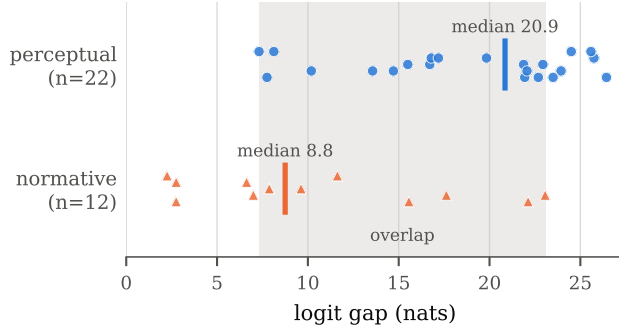


Figure 2: Logit gaps for 22 perceptual decisions and 12 normative ones under the same model and readout. The distributions differ (AUC 0.788, $p = 0.006$) but overlap over most of their range; the shaded band marks the overlap.

Table 2: POPE, 9000 binary presence questions, 1500 yes and 1500 no per split. Recall is identical across splits because the positive half is shared; only the distractors change.

Split	Acc.	Prec.	Rec.	Yes-rate
random	91.2%	98.2%	83.9%	42.7%
popular	89.3%	94.0%	83.9%	44.6%
adversarial	88.3%	92.0%	83.9%	45.6%
all	89.6%	94.7%	83.9%	44.3%

The weakest categories are spatial_relationship (34.5% under CircularEval, against a 25% chance rate) and image_quality (57.3%). The two lowest gaps in our hand-built probe were its spatial-relation item (8.1 nats) and its white-balance item (7.8). A confidence signal read off a handful of examples identified the categories that collapse at scale.

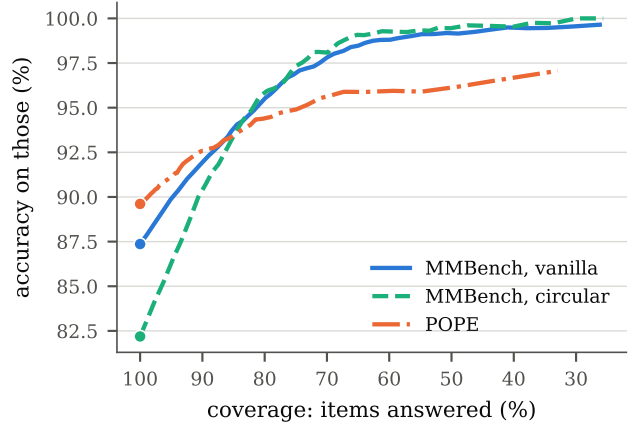


Figure 3: Refusing to answer below a gap threshold. The left-most marker on each curve is the no-abstention operating point; moving right trades coverage for accuracy. MMBench gains far more than POPE, whose binary questions leave the model confidently wrong.

4.6 Binary questions: the signal weakens

MMBench items carry four options. POPE [4] carries two, which is the worst case for a readout that softmaxes over answer slots, and it probes exactly the family—object presence and absence—on which our hand-built probe was perfect.

Accuracy is 89.6% with an F1 of 89.0 (Table 2), but the confidence signal degrades sharply. The gap separates correct from incorrect answers at AUC 0.777, against 0.894 on MMBench, and so, being its monotone image, does the reported probability. The median gap when the model is wrong rises from 5.2 nats on MMBench to 16.4 here: with two slots the model is not merely wrong, it is confidently wrong. The practical consequence is that a threshold tuned on multi-way questions does not transfer to binary ones, and that a semantic if should be given three or more options wherever the decision admits them.

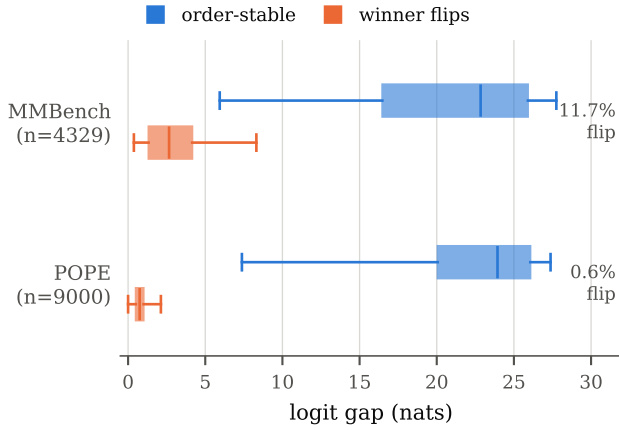


Figure 4: Where option order decides the answer. Boxes span the interquartile range, whiskers the 5th–95th percentiles. Items whose winner flips under rotation sit at gaps an order of magnitude below the stable ones, on both benchmarks.

Table 3: Latency of one decision, by model and placement. CPU figures are float32; thread counts are the best of a sweep.

Model	Device	Threads	ms/forward
Qwen3-VL-4B	RTX 3080	—	110–140
Qwen3-0.6B	i9-12900F	20	340–450
Qwen3.5-4B	i9-12900F	8	1359
Qwen3.5-4B	i9-12900F	20	2392

Order sensitivity moves the other way, and consistently with our account of it: only 0.6% of items change winner under rotation, against 11.7% on MMBench, because binary gaps are large. Those that do flip have a median gap of 0.8 nats, and the gap predicts stability at AUC 0.998.

POPE also shows where object hallucination hides. Its headline metric, the yes-rate, is 42.7–45.6%—below the 50% base rate—so on that metric the model looks clean, and our own prediction that it would over-assert presence was wrong. The effect is real but sits in precision: 98.2% to 94.0% to 92.0% across the random, popular and adversarial splits, while recall stays at exactly 83.9%. Co-occurring distractors do pull the model into false positives; an aggregate yes-rate averages that away.

4.7 Hybrid CPU–GPU execution

Because the two devices are idle at different times, we built a two-stage cascade: the GPU resolves visual predicates over an image, and a CPU model decides an action from the resulting verdicts without ever seeing pixels. Run as two threads with a queue between them, the stages overlap—PyTorch releases the interpreter lock during both CUDA and CPU forwards—and wall-clock time falls from 3.49 s to 2.72 s over four images, a 22% saving.

The thread sweep in Table 3 is a practical result in its own right. On this hybrid CPU, restricting inference to the 8 performance cores is $1.9\times$ faster than using 20 threads; scheduling work onto the efficiency cores actively hurts. Any measurement taken at the default thread count understates CPU inference on such parts, and our own first cas-

cade measurements did.

One optimisation that looks free is not. Questions about one image share a prefix—the system turn and roughly 300 visual tokens—so caching it and running only each question’s tail should save most of the work. The cached path is numerically exact, reproducing every probability to the last bit, and it is 13–14% slower, flat across images of 384, 852 and 1812 tokens with no crossover: it loses the fast attention kernel, which costs more than recomputing the prefix. Image preprocessing accounts for 3% of the time and the forward for 97%.

The policy stage itself did not work, and the way it failed is instructive. Our first design asked one four-option question whose options were not mutually exclusive—an image can need both colour correction and a consent hold. The model oscillated between two *correct* answers, which is a defect of the question, not of the model. Restructuring the stage as independent binary predicates composed in ordinary Python removed the ambiguity. Even then, Qwen3-0.6B answered 1 of 4 policy cases correctly and was unstable under rotation in 3; Qwen3.5-4B answered 3 of 4 and was stable in all 4, but at gaps of 0.7–2.3 nats, below the threshold at which we would trust any of them. The larger model also shifts the pipeline from balanced to 12:1 against the CPU stage, which removes the reason to split the work in the first place.

5 Discussion

The perceptual result is strong and the systems result is modest; the two negative results are what we would want a practitioner to take away.

The first is not a defect of the readout but of the number it hands back. The probability and the gap carry identical information, so nothing is lost in principle; what is lost is precision, and only once the value meets a float32 array. Sixty-two percent of MMBench items land where that array cannot hold the distinction, and the gap that produced them still can. The fix is one line and costs nothing: report the difference of the two logits, not its softmax.

The second is harder to fix and easier to get wrong in production. A semantic *if* will answer a question about your organisation’s consent policy with the same apparent conviction it brings to counting cats, and no signal we measured reliably distinguishes the two. The pattern is therefore safe where the predicate is perceptual and grounded in the input, and unsafe where the predicate is normative. Business policy belongs in code. This is also the lesson of the cascade: the mechanism is called a semantic *if* for a reason, and composing several binary ifs in a host language is both more accurate and more debuggable than asking one model to resolve a multi-way policy.

Limitations. One model, one prompt template, one language, a single machine and a single run, without repetition or confidence intervals on latency. MMBench dev is public and may appear in the model’s training data, so 87.4% is not evidence about generalisation; the claims we defend are about the confidence signal, which is measured within

the same run and does not depend on the absolute score. Our hand-built probe is small and author-annotated, one of its labels was demonstrably wrong before correction, and its 22/22 misled us about the extent of softmax saturation until the larger run corrected it. An earlier draft of this paper also claimed that the gap outperforms the reported probability as a predictor of correctness; computing both AUCs refuted it, and the surviving claim is the narrower one about float32 above. Both errors are the clearest argument in this paper against trusting a small author-written benchmark and an untested intuition. The normative set has 12 items, two of which are arguably entailed by their evidence.

6 Conclusions

Reading a decision from answer-token logits transfers to vision-language models without modification and is accurate and fast on grounded perceptual predicates: 22 of 22 at 110–140 ms per decision on one consumer GPU, robust to option order, and correct on counting, absence, false premises and unanswerable questions. The probability such systems report is the same signal as the logit gap behind it and discriminates no worse; it is simply stored on a scale that float32 cannot hold, collapsing to exactly 1.0 on 62% of items where the gap still ranks them. That signal does not generalise across question families: it fails to separate perceptual questions from normative ones well enough to gate them, and the model is confidently wrong about policy. The signal is also not scale-free: halving the option set from four to two drops its AUC from 0.894 to 0.777 and leaves the model confidently wrong, so a threshold must be set per question shape rather than once. Use semantic ifs for perception, give them three or more options where the decision allows, keep policy in code, and threshold on nats rather than on probabilities.

Code and data

The method is packaged as `discern`, a one-call API in which a below-threshold verdict refuses to be used as a boolean condition. Code, benchmark, raw results and this paper’s sources are available at <https://github.com/XxSamaxX/discern> and archived at <https://doi.org/10.5281/zenodo.22899311>. The scoring engine is vendored from SemIf [3] under the MIT licence.

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